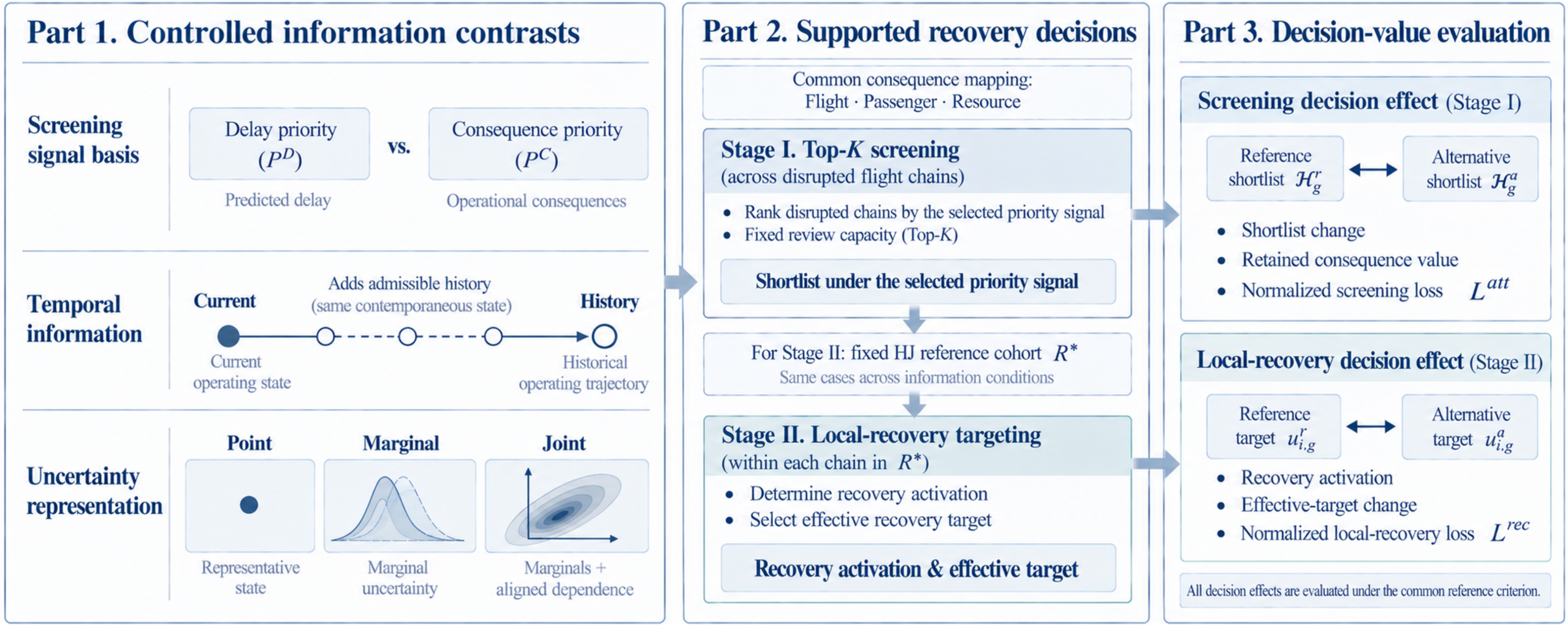

Part 1. Controlled information contrasts
Screening signal basis
Delay priority (P^D)
vs.
Consequence priority (P^C)
Predicted delay
Operational consequences
Temporal information
Current
Adds admissible history
(same contemporaneous state)
History
Current operating state
Historical operating trajectory
Uncertainty representation
Point
Marginal
Joint
Representative state
Marginal uncertainty
Marginals + aligned dependence
Part 2. Supported recovery decisions
Common consequence mapping:
Flight · Passenger · Resource
Stage I. Top-K screening
(across disrupted flight chains)
Rank disrupted chains by the selected priority signal
Fixed review capacity (Top-K)
Shortlist under the selected priority signal
For Stage II: fixed HJ reference cohort R^*
Same cases across information conditions
Stage II. Local-recovery targeting
(within each chain in R^*)
Determine recovery activation
Select effective recovery target
Recovery activation & effective target
Part 3. Decision-value evaluation
Screening decision effect (Stage I)
Reference shortlist $\mathcal{H}_g^r$
Alternative shortlist $\mathcal{H}_g^a$
Shortlist change
Retained consequence value
Normalized screening loss L^{att}
Local-recovery decision effect (Stage II)
Reference target $u_{i,g}^r$
Alternative target $u_{i,g}^a$
Recovery activation
Effective-target change
Normalized local-recovery loss L^{rec}
All decision effects are evaluated under the common reference criterion.